

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 04

DATE : 05-01-2020

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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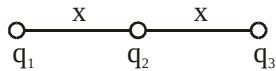
Your Target is to secure Good Rank in NEET 2020

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. Three charged particles are placed on a straight line as shown in Fig. q_1 and q_2 are fixed, but q_3 can be moved. Under the action of the forces from q_1 and q_2 , q_3 is in equilibrium. What is the relation between q_1 and q_2 ?

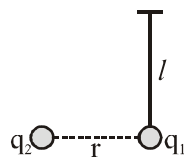


- (1) $q_1 = 4q_2$ (2) $q_1 = -q_2$
(3) $q_1 = -4q_2$ (4) $q_1 = q_2$
2. Two particles A and B (B is right of A) having charges 8×10^{-6} C and -2×10^{-6} C, respectively, are held fixed with separation of 20 cm. Where should a third charged particle be placed so that it does not experience a net electric force :-

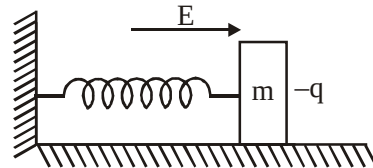
- (1) 5 cm right of B
(2) 5 cm left of A
(3) 20 cm left of A
(4) 20 cm right of B

3. An isolated charge q_1 of mass m is suspended freely by a thread of length l . Another charge q_2 is brought near it ($r \gg l$). When q_1 is in equilibrium, tension in thread will be :-

- (1) mg
(2) $>mg$
(3) $<mg$
(4) none of these



4. A block of mass m containing a net negative charge $-q$ is placed on a frictionless horizontal table and is connected to a wall through an unstretched spring of spring constant k . If the horizontal electric field E parallel to the spring is switched on, then the maximum compression of the spring is :-



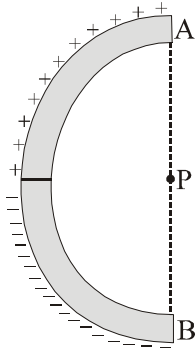
- (1) $\sqrt{qE/k}$ (2) $2qE/k$
(3) qE/k (4) zero
5. Five point charges, each of value $+q$, are placed on five vertices of a regular hexagon of side L . The magnitude of the force on a point charge of value $-q$ coulomb placed at the center of the hexagon is

- (1) $\frac{1}{\pi\epsilon_0} \left(\frac{q}{L} \right)^2$
(2) $\frac{2}{\pi\epsilon_0} \left(\frac{q}{L} \right)^2$
(3) $\frac{1}{2\pi\epsilon_0} \left(\frac{q}{L} \right)^2$
(4) $\frac{1}{4\pi\epsilon_0} \left(\frac{q}{L} \right)^2$

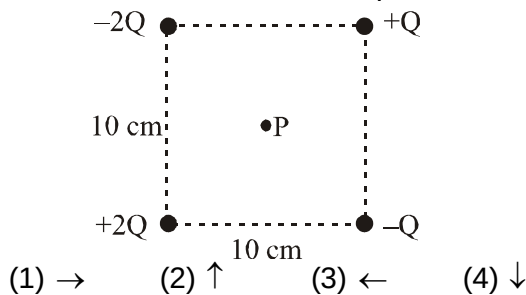
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6. A thin glass rod is bent into a semicircle of radius r . A charge $+Q$ is uniformly distributed along the upper half, and a charge $-Q$ is uniformly distributed along the lower half, as shown in Fig. The electric field E at P , the center of the semicircle, is :-

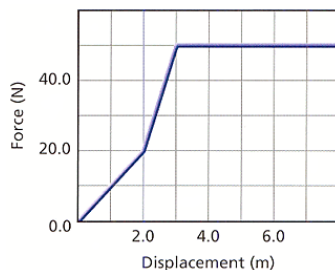
- (1) $\frac{Q}{\pi^2 \epsilon_0 r^2}$
(2) $\frac{2Q}{\pi^2 \epsilon_0 r^2}$
(3) $\frac{4Q}{\pi^2 \epsilon_0 r^2}$
(4) $\frac{Q}{4\pi^2 \epsilon_0 r^2}$



7. Four electrical charges are arranged on the corners of a 10 cm square as shown. What would be the direction of the resulting electric field at the center point P ?



- 8.



Work Done by force from $x = 2\text{m}$ to $x = 4\text{m}$ is

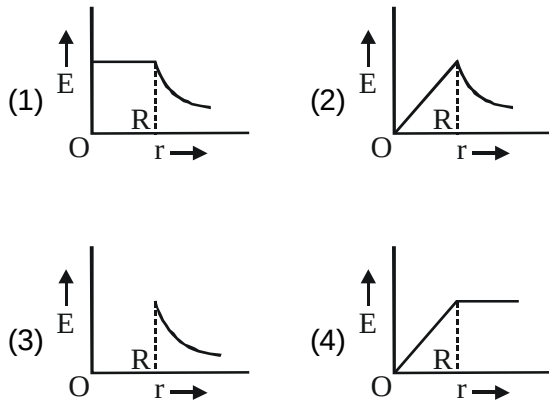
- (1) 140 J (2) 70 J (3) 35 J (4) 85 J

9. Two equal negative charges each $-q$, are placed at points $(0, a)$ and $(0, -a)$ on y -axis. A positive charge q is released from point $(2a, 0)$. This charge will be :-
(1) execute SHM about the origin
(2) oscillate but not execute SHM
(3) move towards origin and will become stationary
(4) execute SHM along y -axis
10. The net electric flux ϕ through a hemisphere surface of radius R , placed in a uniform electric field of intensity E which is parallel to the axis of its circular plane is :-
(1) $2\pi RE$ (2) $2\pi R^2 E$
(3) Zero (4) $(4/3)\pi R^3 E$
11. Mark the correct statement about the electric lines of force :-
(1) The number of lines of force leaving a +ve charge or entering a -ve charge is proportional to the magnitude of charge
(2) The path of a charged particle in an electric field is not always along the field lines
(3) The direction of electric lines of force is along the normal to the conducting surfaces
(4) All of the above
12. A circular ring of radius R which has uniformly distributed charge Q . Find the flux crossing through a sphere of radius R having its centre on the periphery of the ring :-

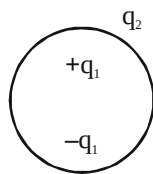
- (1) zero (2) $\frac{Q}{4\epsilon_0}$ (3) $\frac{Q}{\epsilon_0}$ (4) $\frac{Q}{3\epsilon_0}$

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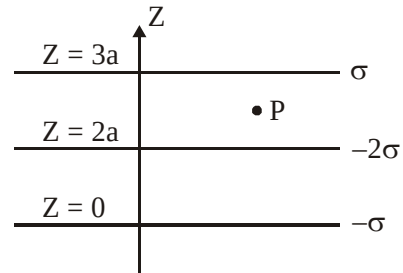
13. The electric field due to a uniformly charged non conducting sphere of radius R as a function of the distance from its centre is represented graphically by :-



14. Consider the charge configuration and a spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the electric field will be due to:-



- (1) q_2
(2) only the positive charges
(3) all the charges
(4) $+q_1$ and $-q_1$
15. Three infinitely charged non conducting sheets are kept parallel to XY-plane having charge densities as shown in figure. Find out the net electric field at point P :-

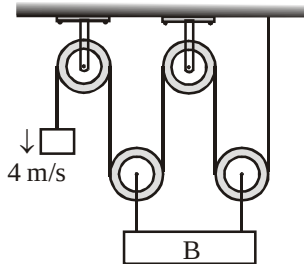


- (1) $\frac{2\sigma}{\epsilon_0} \hat{k}$ (2) $-\frac{2\sigma}{\epsilon_0} \hat{k}$
(3) $-\frac{4\sigma}{\epsilon_0} \hat{k}$ (4) $\frac{4\sigma}{\epsilon_0} \hat{k}$
16. Positive and negative point charges of equal magnitude are kept at $\left(0, 0, \frac{a}{2}\right)$ and $\left(0, 0, -\frac{a}{2}\right)$ respectively. The work done by the electric field when another positive point charge is moved from $(-a, 0, 0)$ to $(0, a, 0)$ is :-
(1) positive
(2) negative
(3) zero
(4) dependent on the path connecting the initial and final positions
17. An electron (of charge $-e$) revolves around a long wire with uniform charge density λ in a circular path of radius r . Its kinetic energy is given by :-

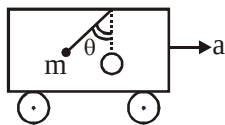
- (1) $\frac{\lambda e}{2\pi \epsilon_0 r}$ (2) $\frac{\lambda e}{4\pi \epsilon_0 r}$
(3) $\frac{\lambda e}{2\pi \epsilon_0}$ (4) $\frac{\lambda e}{4\pi \epsilon_0}$

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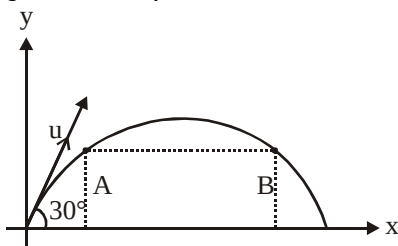
18. For the given system find the velocity of block 'B' if velocity of block 'A' is 4 m/s in downward direction :- (Strings & pulley are ideal)



- (1) $V_B = 1 \text{ m/s} \uparrow$ (2) $V_B = 2 \text{ m/s} \downarrow$
(3) $V_B = 4 \text{ m/s} \uparrow$ (4) $V_B = \text{None}$
19. If trolley accelerates horizontally with acceleration a then bob is displaced backward from its initial vertical position. The angular deflection of the bob in equilibrium is :-

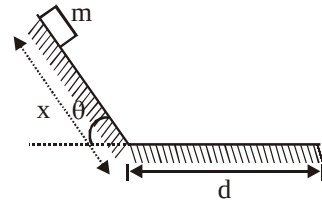


- (1) $\theta = \cos^{-1} \left(\frac{a}{g} \right)$ (2) $\theta = \sin^{-1} \left(\frac{a}{g} \right)$
(3) $\theta = \cot^{-1} \left(\frac{a}{g} \right)$ (4) $\theta = \tan^{-1} \left(\frac{a}{g} \right)$
20. A particle is thrown with a velocity of $v \text{ m/s}$. it passes A and B as shown in figure at time $t_1 = 1 \text{ s}$ and $t_2 = 3 \text{ s}$. The value of u is ($g = 10 \text{ m/s}^2$) :-



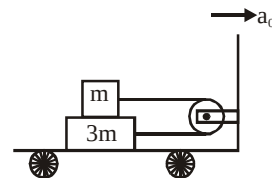
- (1) 20 m/s (2) 10 m/s
(3) 40 m/s (4) 5 m/s

21. A block of mass m is released on the top of a smooth inclined plane of length x and inclination θ as shown in figure. Horizontal surface is rough. If block comes to rest after moving a distance d on the horizontal surface, then coefficient of friction between block and surface is :-



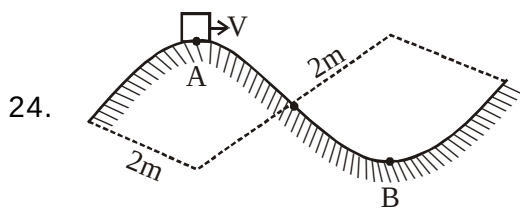
- (1) $\frac{x \sin \theta}{2d}$ (2) $\frac{x \cos \theta}{2d}$
(3) $\frac{x \sin \theta}{d}$ (4) $\frac{x \cos \theta}{d}$

22. In the given figure if all the surfaces are smooth then the acceleration of the two boxes relative to the trolley is.



- (1) $3a_0/2$ (2) a_0
(3) $a_0/2$ (4) $2a_0$
23. Which of the following cannot be the safe speed of a car at a turn on a flat horizontal road of radius 120 m and friction coefficient 0.75.
- (1) 72 km/hr (2) 108 km/hr
(3) 110 km/hr (4) 30 km/hr

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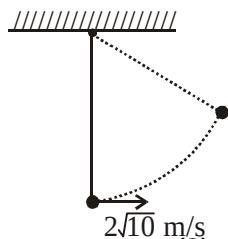
If a block of mass 100 kg passes from point A and B with same speed and it is found that difference between normal reactions at point B and point A is 400 N then find speed of car at point B.

- (1) 2 m/s (2) 4 m/s
(3) 20 m/s (4) None of these

25. If a particles position vectors is $\vec{r} = 2\cos\theta\hat{i} + 2\sin\theta\hat{j}$. If particles speed at each instant is 4 m/s then find acceleration of particle.

- (1) 4 (2) 18
(3) 16 (4) 8

26. If a bob is given a velocity of $2\sqrt{10}$ m/s at lowest position then find tension in the string at extreme position. If mass of bob is 5 kg and string is 10m long.

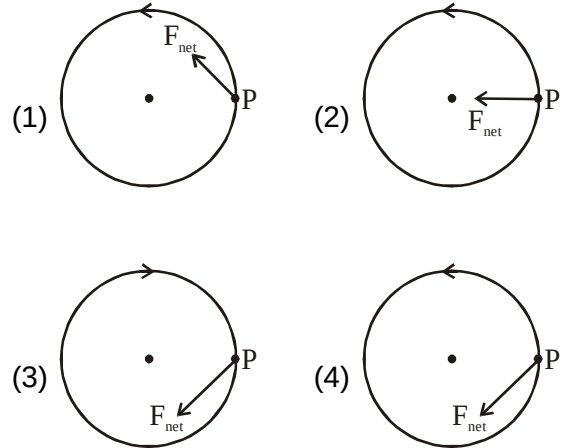


- (1) 20 N (2) 40 N
(3) 60 N (4) 10 N

27. Find speed of a car on a banked road of angle 60° so that friction is not used if radius of track is $270\sqrt{3}$ m.

- (1) 81 m/s (2) 90 m/s
(3) 30 m/s (4) None

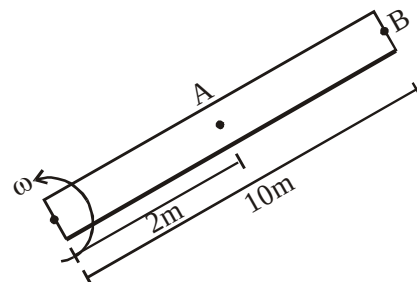
28. What can be the representation of net force on a particle (P) which is slowing down with a constant rate in circular motion.



29. If a particle at rest is subjected to angular acceleration $\alpha = (2t + 3t^2 + 2)$ (in rad/s²) find its angular velocity (in rad/s) after 2 seconds.

- (1) 4 (2) 16 (3) 8 (4) 12

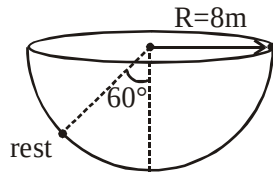
30. If a rod of length 10 m is rotating about a fixed point as shown in a horizontal plane with constant angular velocity of 5 rad/sec. then find ratio of linear velocities of point A to point B.



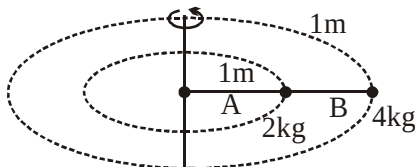
- (1) 10 (2) 5 (3) 0.2 (4) 0.1

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31. A ball is released from rest as shown in a frictionless hemispherical bowl. If mass of ball is 2 kg then find normal reaction when ball reaches lowest point.



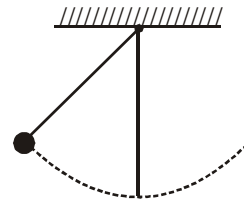
- (1) 20N (2) 40N
(3) 60N (4) 80N
32. If two particles are rotated in horizontal circle as shown if at an instant velocity of 4 kg is 4 m/s find ratio of tension in string A to the tension in string B.



- (1) $\frac{4}{5}$ (2) $\frac{5}{4}$
(3) $\frac{1}{4}$ (4) 4
33. For a particle starting from rest under constant angular acceleration of 2 rad/sec^2 . In a circle of radius 1m. Find magnitude of net acceleration in m/s^2 after 2 sec.

- (1) $\sqrt{260}$ (2) 4
(3) $\sqrt{20}$ (4) $\sqrt{256}$

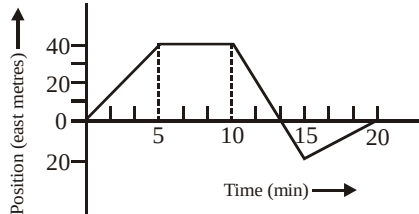
34. A pendulum bob is released from rest as shown then find incorrect statement.



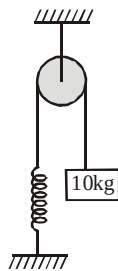
- (1) work done by tension increases kinetic energy as it comes down.
(2) gravitational potential energy changes as bob moves.
(3) tension increases as bob moves in downward direction.
(4) velocity of bob is maximum at lowest position.
35. A particle is projected at 60° to the horizontal with a kinetic energy K. The kinetic energy at the highest point is :-
(1) K (2) 0
(3) $\frac{K}{4}$ (4) $\frac{K}{2}$
36. A swimmer can swim in still water with 5 m/s. If a river is flowing with 4 m/s then find the time to cross the river of width 300 m if he has to reach to exactly opposite point :-
(1) 60 sec.
(2) 75 sec.
(3) 100 sec.
(4) $\frac{100}{3}$ sec.

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37. A boy begins to walk eastward along a street in front of his house and the graph of his position from home is shown in the following figure. His average speed for the whole time interval is equal to :-

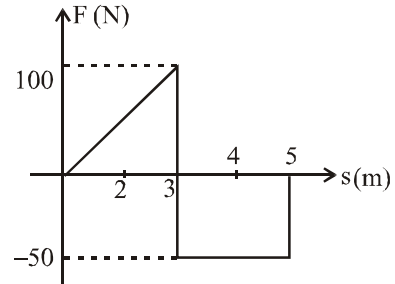


- (1) 8 m/min (2) 6 m/min
(3) $\frac{8}{3}$ m/min (4) 2 m/min
38. The mass of the bob of a simple pendulum is m and it is raised to a maximum height h . The work done by gravity in moving it from one extreme end to the other is
(1) $2 mgh$ (2) mgh
(3) $3 mgh$ (4) zero
39. Two spring A and B have spring constants K_A and K_B such that $K_B = 2 K_A$. The free ends of both the springs are stretched by same force. If energy stored in spring A is E then energy stored in spring B will be
(1) $E/2$ (2) $2E$ (3) E (4) $4E$
40. For the system shown in figure the potential energy stored in the spring is ($K = 200$ N/m)



- (1) 10 J (2) 15 J (3) 25 J (4) 50 J

41. A time varying power $P = 2t$ is applied on a particle of mass 3kg. The kinetic energy of the particle at $t = 4$ s is
(1) 4 J (2) 8 J (3) 16 J (4) 24 J
42. In the given graph if mass of body $m = 10$ kg and initial speed is 10 m/s. Then find out KE of body after a displacement of 5 m :-



- (1) 50 J (2) 400 J
(3) 550 J (4) 500 J
43. If K.E. of a body is decreased by 36%. Then % change in momentum is :-
(1) 20% ↓ (2) 20% ↑
(3) 40% ↓ (4) 40% ↑
44. If P.E. of two molecules is given as

$U = \frac{A}{r^{10}} - \frac{B}{r^5}$ where A and B are constant and 'r' is separation between molecules then find equilibrium position in terms of r .

- (1) $\left(\frac{2A}{B}\right)^{-1/6}$ (2) $\left(\frac{A}{2B}\right)^{-1/5}$
(3) $\left(\frac{A}{B}\right)^{1/6}$ (4) $\left(\frac{2A}{B}\right)^{1/5}$

45. A fan keeps working for 30 minutes and uses energy of 90000 J, power consumed by fan is
(1) 3000 W (2) 50 W
(3) 500 W (4) 300 W

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46. In which of the following cases, entropy of the system is increasing.
- (1) $I_2(g) \longrightarrow I_2(s)$
 - (2) $C_{(graphite)} \longrightarrow C_{(diamond)}$
 - (3) Ideal monoatomic gas subjected to change on state from 2 atm 300 K to 2 atm 600 K.
 - (4) Dimerisation of acetic acid.
47. The entropy change when two moles of ideal monoatomic gas is heated from 200°C to 300°C reversibly and isochorically?
- (1) $\frac{3}{2}R \ln\left(\frac{300}{200}\right)$
 - (2) $\frac{5}{2}R \ln\left(\frac{573}{273}\right)$
 - (3) $3R \ln\left(\frac{573}{473}\right)$
 - (4) $\frac{3}{2}R \ln\left(\frac{573}{473}\right)$
48. For the gas-phase decomposition,
- $$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$
- (1) $\Delta H < 0, \Delta S < 0$
 - (2) $\Delta H > 0, \Delta S > 0$
 - (3) $\Delta H > 0, \Delta S < 0$
 - (4) $\Delta H < 0, \Delta S > 0$
49. $\Delta H = 30 \text{ kJ mol}^{-1}$, $\Delta S = 75 \text{ J/K/mol}$. Find boiling temperature at 1 atm :
- (1) 400 K
 - (2) 300 K
 - (3) 150 K
 - (4) 425 K
50. For the process
 $H_2O(l)(1 \text{ bar}, 373 \text{ K}) \longrightarrow H_2O(g)(1 \text{ bar}, 373 \text{ K})$,
the correct set of thermodynamic parameters is :
- (1) $\Delta G = 0, \Delta S = +ve$
 - (2) $\Delta G = 0, \Delta S = -ve$
 - (3) $\Delta G = +ve, \Delta S = 0$
 - (4) $\Delta G = -ve, \Delta S = +ve$
51. What are the signs for ΔH , ΔS and ΔG for the freezing of liquid water at -10°C ?
- | | ΔH | ΔS | ΔG |
|-----|------------|------------|------------|
| (1) | + | - | + |
| (2) | - | - | 0 |
| (3) | - | + | - |
| (4) | - | - | - |
52. "A perfect crystalline substance has an entropy of zero at absolute zero" is a statement of :-
- (1) Hess' Law.
 - (2) The First Law of Thermodynamics.
 - (3) The Second Law of Thermodynamics.
 - (4) The Third Law of Thermodynamics.
53. Select the option in which all the parameters are intensive :
- (1) pH, volume, Electrode potential.
 - (2) Molar enthalpy, Heat capacity, Resistivity.
 - (3) Temperature, Molality, Electromotive force.
 - (4) Mass, Pressure, Molar entropy.
54. Calculate ΔH when 2 moles of solid benzoic acid undergo complete combustion at 300K if
- $$C_6H_5COOH(s) + \frac{15}{2}O_2(g) \longrightarrow 7CO_2(g) + 3H_2O(l)$$
- $\Delta U_{\text{reaction}} = -750 \text{ kJ/mole}$
- (1) - 751.247 kJ
 - (2) - 752.494 kJ
 - (3) - 1501.247 kJ
 - (4) - 1502.494 kJ

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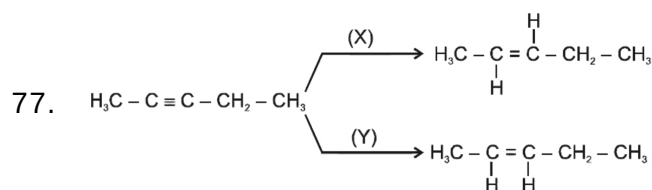
55. An ideal gas is expanded irreversibly against 10 bar pressure from 20 litre to 30 litre. Calculate q if process is isothermal :
- (1) 0 (2) + 100 J
(3) - 100 J (4) + 10 kJ
56. Warming ammonium chloride with sodium hydroxide in a test tube is an example of :
- (1) closed system
(2) isolated system
(3) open system
(4) none of these
57. In thermodynamics, a process is called reversible when:
- (1) surroundings and system change into each other
(2) there is no boundary between system and surroundings
(3) the surroundings are always in equilibrium with the system
(4) the system changes into the surroundings spontaneously
58. For which of the following change $\Delta H \neq \Delta E$?
- (1) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \longrightarrow 2\text{HI}(\text{g})$
(2) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \longrightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
(3) $\text{C}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$
(4) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$
59. A system is provided 50 J of heat and work done on the system is 10 J. The change in internal energy during the process is :
- (1) 40 J (2) 60 J
(3) 80 J (4) 50 J
60. Magnitude of enthalpy of neutralization is minimum for
- (1) $\text{HCN} + \text{KOH}$
(2) $\text{HCl} + \text{KOH}$
(3) $\text{HCl} + \text{NH}_4\text{OH}$
(4) $\text{HCN} + \text{NH}_4\text{OH}$
61. An athlete is given 100 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) of energy equivalent to 1560 kJ. He utilizes 50% of this gained energy in the event. In order to avoid storage of energy in the body, the weight of water he would need to perspire is :
(The enthalpy of evaporation of water is 44 kJ/mole.)
- (1) 319 g (2) 422 g
(3) 293 g (4) 378 g
62. AB , A_2 and B_2 are diatomic molecules. If the bond enthalpies of A_2 , AB and B_2 are in the ratio 1:1:0.5 and enthalpy of formation of AB from A_2 and B_2 is -100 kJ/mol^{-1} . What is the bond enthalpy of A_2 ?
- (1) 400 kJ/mol
(2) 200 kJ/mol
(3) 100 kJ/mol
(4) 300 kJ/mol
63. Which of the following equation gives the values of heat of formation (ΔH_f°) ?
- (1) $\text{C}(\text{diamond}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$
(2) $\frac{1}{2} \text{H}_2(\text{g}) + \frac{1}{2} \text{F}_2(\text{g}) \longrightarrow \text{HF}(\text{g})$
(3) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$
(4) $\text{H}_2(\text{g}) + \text{F}_2(\text{g}) \longrightarrow 2\text{HF}(\text{g})$

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64. If x_1 , x_2 and x_3 are enthalpies of H–H, O=O and O–H bonds respectively, and x_4 is the enthalpy of vaporisation of water, estimate the standard enthalpy of combustion of hydrogen
- (1) $x_1 + \frac{x_2}{2} - 2x_3 + x_4$
- (2) $x_1 + \frac{x_2}{2} - 2x_3 - x_4$
- (3) $x_1 + \frac{x_2}{2} - x_3 + x_4$
- (4) $2x_3 - x_1 - \frac{x_2}{2} - x_4$
65. Oxidation of isobutylene with acidic potassium permanganate gives
- (1) Acetone + CO_2
- (2) Acetic acid
- (3) Acetic acid + CO_2
- (4) Acetic acid + acetone
66. Which of the following is maximum reactive for NBS
- (1) PhCH_3
- (2) $\text{PhCH}_2\text{—CH}_3$
- (3) $\text{PhCH}_2\text{—CH=CH}_2$
- (4) Ph—CH—CH=CH_2
 CH_3
67. The correct order of increasing stability of the given alkenes is
- (1) 1 – pentene > cis – 2 – pentene > trans – 2 – pentene > 2 – methyl – 2 – butene
- (2) 1 – pentene > trans – 2 – pentene > cis – 2 – pentene > 2 – methyl – 2 – butene
- (3) 1 – pentene < cis – 2 – pentene < trans – 2 – pentene < 2 – methyl – 2 – butene
- (4) 1 – pentene < trans – 2 – pentene < cis – 2 – pentene < 2 – methyl – 2 – butene
68. The Corey-House alkane synthesis is carried out by treating an alkyl halide with
- (1) Lithium metal
- (2) Copper metal
- (3) Lithium metal followed by reaction with cuprous iodide and then treating the product with an alkyl halide
- (4) Cuprous iodide followed by reaction with alkyl halide
69. Which of the following acids on decarboxylation gives isobutane
- (1) 2,2–Dimethyl butanoic acid
- (2) 2,2–dimethyl propanoic acid
- (3) 3–Methyl pentanoic acid
- (4) 2–Methyl butanoic acid
70. Which alkane is liquid at room temperature
- (1) Hexane
- (2) Butane
- (3) Propane
- (4) Ethane

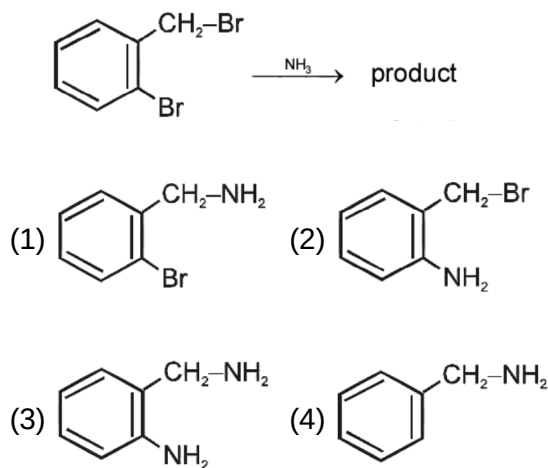
SPACE FOR ROUGH WORK

71. The compound with the highest boiling point is
 (1) n-hexane
 (2) n-pentane
 (3) 2,2-dimethyl propane
 (4) propane
72. The pH of solution in Kolbe's electrolysis
 (1) increase with time
 (2) decrease with time
 (3) remains constant
 (4) nothing can be said
73. Clemmensen reduction is not preferred for compounds which have _____ group.
 (1) Base sensitive
 (2) Acid sensitive
 (3) both of the above
 (4) none of these
74. On oxidation, an unknown alkene C_8H_{16} gives a mixture of propanoic acid and pentanoic acid. The unknown alkene can be
 (1) cis-3-octene
 (2) trans-2-octene
 (3) cis-4-octene
 (4) trans-4-octene
75. Acetone will be obtained on ozonolysis of
 (1) 1-pentene (2) 2-pentene
 (3) isopentene (4) 2-pentyne
76. Baeyer's reagent is :-
 (1) alkaline $KMnO_4$ solution
 (2) acidic $KMnO_4$ solution
 (3) neutral $KMnO_4$ solution
 (4) aqueous bromine solution

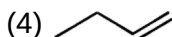
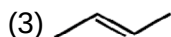
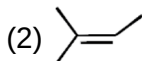
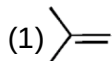
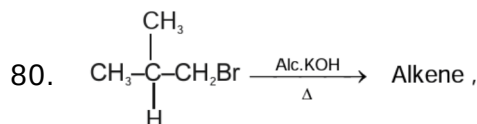


In the above reaction the using reagents X and Y are :-

- (1) Na/ liq. NH_3 for X
 (2) H_2 , Pd/ $BaSO_4$ for Y
 (3) BH_3 -THF + CH_3COOH for Y
 (4) All of these
78. S_N2 mechanism proceeds through intervention of :-
 (1) Carbonium ion
 (2) Transition state
 (3) Free radical
 (4) Carbanion
79. What is the major product obtained in the following reaction ?



SPACE FOR ROUGH WORK



81. Density of water is 1g/mL. The concentration of water in mol/litre is-

- (1) 1000 (2) 18
(3) 0.018 (4) 55.5

82. 1000 gram aqueous solution of CaCO_3 contains 10 gram of CaCO_3 . Concentration of solution is-

- (1) 10 ppm (2) 100 ppm
(3) 1000 ppm (4) 10,000 ppm

83. Equal volumes of 0.1 M AgNO_3 and 0.2 M NaCl are mixed. The concentration of NO_3^- ions in the mixture will be-

- (1) 0.1 M (2) 0.05 M
(3) 0.2 M (4) 0.15 M

84. 10 gram of glucose are dissolved in 150 gram of water. The mass % of glucose is-

- (1) 5% (2) 6.25%
(3) 93.75% (4) 15%

85. Vapour pressure of a solution containing nonvolatile solute is -

- (1) More than the vapour pressure of a solvent
(2) Less than the vapour pressure of solvent
(3) Equal to the vapour pressure of solvent
(4) None of these

86. In cold countries, ethylene glycol is added to water in the radiators of cars during winters. It results in-

- (1) Lowering in boiling point
(2) Reducing viscosity
(3) Reducing specific heat
(4) Lowering in freezing point

87. An example of colligative property is-

- (1) Freezing point
(2) Boiling point
(3) Vapour pressure
(4) Osmotic pressure

88. Which solution will show maximum elevation in b.pt-

- (1) 0.1 M KCl
(2) 0.1 M BaCl_2
(3) 0.1 M FeCl_3
(4) 0.1 M $\text{Fe}_2(\text{SO}_4)_3$

89. 0.5 M solution of urea is isotonic with-

- (1) 0.5 M NaCl solution
(2) 0.5 M sugar solution
(3) 0.5 M BaCl_2 solution
(4) 0.5 M solution benzoic acid in benzene

90. Which one of the following salts would have the same value of the Vant Hoff factor (i) as that of $\text{K}_3[\text{Fe}(\text{CN})_6]$.

- (1) NaCl
(2) Na_2SO_4
(3) $\text{Al}_2(\text{SO}_4)_3$
(4) $\text{Al}(\text{NO}_3)_3$

SPACE FOR ROUGH WORK

91. Mendel conducted hybridisation experiments on garden peas from:
(1) 1842-1847
(2) 1856-1863
(3) 1863-1873
(4) 1820-1831
92. The phenotypic ratio of a monohybrid cross in F_2 generation is:
(1) 3 : 1 (2) 1 : 2 : 1
(3) 2 : 1 : 1 (4) 1 : 3 : 1
93. Which of the following trait of garden pea is not present on 4th chromosome?
(1) Pod shape
(2) Flower position
(3) Flower colour
(4) Stem height
94. Which of the following genotype represents heterozygous condition?
(1) AABB (2) aaBB
(3) AaBb (4) aabb
95. Who performed hybridisation experiments on garden pea before Mendel?
(1) Goss and Knight
(2) Darwin
(3) Hugo de Vries
(4) Correns
96. When two hybrids are crossed, the percentage of dominant is:
(1) 75 % (2) 25 %
(3) 50% (4) 100%
97. Types of gametes produced by an individual heterozygous for three genes is :
(1) 8 (2) 6
(3) 9 (4) 16
98. Type of gametes formed by the plant with genotype Aa bb CC Dd:
(1) 4 (2) 8
(3) 16 (4) 32
99. How many zygotic combinations are possible between a cross AA BB Cc Dd × aa Bb Cc Dd?
(1) 32 (2) 64
(3) 16 (4) 18
100. A dwarf pea plant is treated with gibberellin which makes it tall. This plant is crossed with dwarf pea plant. The phenotypic ratio in next generation shall be :
(1) 3 tall: 1 dwarf
(2) 50 % tall: 50 % dwarf
(3) All tall
(4) All dwarf
101. If an organism produces 32 types of gametes. Its genotype should be heterozygous for:
(1) 4 genes (2) 5 genes
(3) 8 genes (4) 2 genes
102. Skin colour in human, is controlled by three gene pairs, then what is the number of intermediate & parental combinations in F_2 generation respectively :-
(1) 2, 62 (2) 62, 2
(3) 2, 14 (4) 14, 2
103. Which of the following type of inheritance can we find out with the help of reciprocal cross?
(1) Cytoplasmic inheritance
(2) Gene inheritance
(3) Maternal inheritance
(4) (1) & (3) both
104. Genes a, b and c assort independently and are recessive to their respective alleles A, B and C. Two triple heterozygous AaBbCc individual are crossed what is the probability that a given offspring will be genotypically homozygous for all three dominant alleles?
(1) 2/64 (2) 27/64
(3) 1/64 (4) 6/64
105. A dihybrid plant is with genotype GgTt. In this plant first gene pair is codominant on selfing of this plant what is phenotypic ratio?
(1) 2 : 1
(2) 3 : 1
(3) 3 : 6 : 3 : 1 : 2 : 1
(4) 6 : 3 : 2 : 1

106. Which will not the genotype of pollen grain of a plant with genotype HHbbDdnnMaa?
- HbdNMa
 - HbDnMa
 - HbdnMa
 - All the above
107. The allele, which is unable to express its effect in the presence of another is called :-
- Co-dominant
 - Supplementary
 - Complementary
 - Recessive
108. In the inheritance of flower colour in dog flower plant, the F_1 had a phenotype that
- Resembles both of the parents
 - Did not resemble either of the two parents
 - Resemble with only one parent
 - (1) & (3) both
109. The three different alleles of human ABO blood types will produce how many genotypes & phenotypes respectively :-
- 4 & 6
 - 6 & 4
 - 6 & 6
 - 4 & 4
110. Other than pea plants it was found that sometimes the F_1 had a phenotype that did not resemble either of the two parents and was in between the two. Due to :-
- Complete dominance
 - Incomplete dominance
 - Co-dominance
 - Complementary gene interaction
111. Which of the following condition is true for co-dominance :-
- Phenotype of F_1 resembled either of the two parents
 - Phenotype of F_1 did not resemble either of two parents
 - Phenotype of F_1 resembles both parents
 - None of these

112. Which of the following is a good example of multiple allele :-
- ABO blood group
 - Size of starch grain in pea
 - Shape of seed
 - Flower colour in pea
113. In mice, black coat colour (allele B) is dominant to brown coat colour (allele b). The offspring of a cross between a black mouse (BB) and a brown mouse (bb) were allowed to interbreed. What percentage of the progeny would have black coats ?
- 25 %
 - 50 %
 - 75 %
 - 100 %
114. Fruit shape in shepherd's purse (*Capsella bursa*) is of two types—triangular and top-shaped. Triangular fruit shape (T) is dominant over top-shape (t). Following table summarizes the results of several crosses.

Cross	Result
Strain 1 $\times$ tt	All triangular
Strain 2 $\times$ tt	1 triangular : 1 top – shaped
Strain 3 $\times$ tt	All top – shaped
Strain 4 $\times$ Tt	3 triangular : 1 top – shaped

Which pair of strains possess the genotype Tt?

- Strains 2 and 3
 - Strains 2 and 4
 - Strains 1 and 3
 - Strains 1 and 4
115. Which one of the following crosses would have 1 : 1 : 1 : 1 ratio ?
- TtRR $\times$ ttrr
 - TTRR $\times$ ttrr
 - TtRr $\times$ ttrr
 - TtRR $\times$ TTrr
116. Possible blood group in children from the parents with 'B' and 'O' blood group are :
- All B
 - All O
 - Both B and O
 - A and B

117. Reciprocal cross is

- (1) Intraspecific hybridization
- (2) Back cross
- (3) Pollen grains from one variety deposited on the stigma of the contrasting variety and vice-versa, in hybridization experiments
- (4) Test cross

118. During secondary growth in dicot stem, vascular cambium ring is formed by intrafascicular cambium and:

- (1) Cork
- (2) Phellogen
- (3) Interfascicular cambium
- (4) Phellem

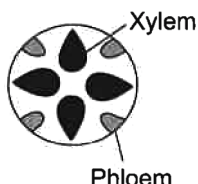
119. In epidermal tissue system, what is correct about an epidermal cell?

- (1) Collenchymatous cell with abundant cytoplasm
- (2) Collenchymatous cell with little cytoplasm
- (3) Parenchymatous cell with abundant cytoplasm
- (4) Parenchymatous cell with little cytoplasm

120. Select the incorrect match.

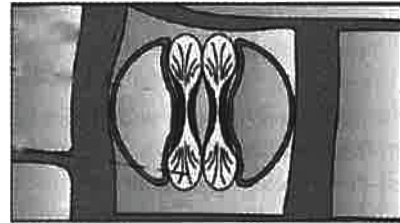
- (1) Sieve element - Conduction of photosynthetic
- (2) Bast fibre - Support
- (3) Ray parenchyma - Water storage
- (4) Phloem parenchyma - Food storage

121. The given vascular bundle is present in



- (1) Dicot steam
- (2) Dicot root
- (3) Radish
- (4) Isobilateral leaf

122. Identify the following figure and choose correct option.



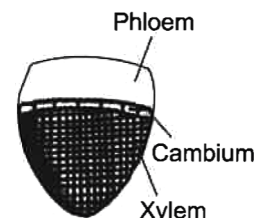
- (1) Sunken stomata present in xerophytic plants
- (2) Dumb-bell shaped guard cell present in grasses
- (3) Scotoactive stomata of succulent xerophytic
- (4) Stomata with bean-shaped guard cells

123. Select incorrect match pair.—

- (1) Open vascular bundles - Cambium present
- (2) Stomata - Transpiration and gaseous exchange
- (3) Protophloem - Derived from vascular cambium
- (4) Bicollateral vascular bundle - Cucurbita stem

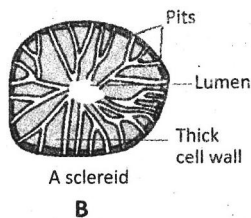
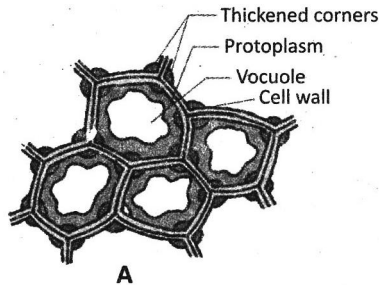
124. In the given figure, vascular bundle is

- (A) Radial
- (B) Conjoint
- (C) Collateral
- (D) Bicollateral
- (E) Open
- (F) Closed
- (G) Concentric



- (1) A, C and F
- (2) B, D and E
- (3) B, C and E
- (4) B, C and F

125. Diagram of some simple permanent tissues are given (A and B). Select the option with correct information about A and B :

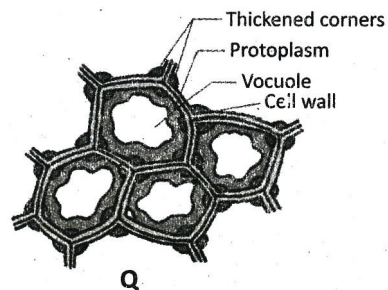
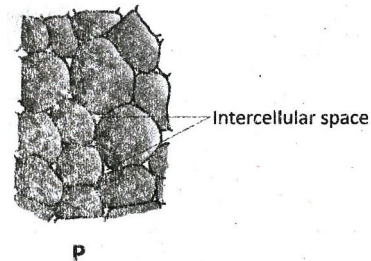


	A	B
(1)	Dead cells	Living cells
(2)	Provide mechanical strength	Mainly photosynthetic in action
(3)	Devoid of protoplasm	Lignified cell wall
(4)	Cell wall thickened at corners	Protoplasm absent

126. Which of the following provides mechanical support to the growing parts of plants like young stem and petiole of a leaf?

- (1) Sclerenchyma
- (2) Collenchyma
- (3) Parenchyma
- (4) Xylem vessels

127. P and Q are two types of simple permanent tissues. Select the correct option with their occurrence :



	P	Q
(1)	Cortex of dicot stem	Below epidermis in dicot stem
(2)	Pulp of guava	Pith in monocot root
(3)	Medullary rays	Epidermis of stem
(4)	Xylem and phloem	Lignified fibres

128. Sclereids are found in all of the following except :

- (1) Solid endosperm of coconut
- (2) Fruit wall of nuts
- (3) Seed coat of legumes
- (4) Pulp of fruits like pear, guava and sapota

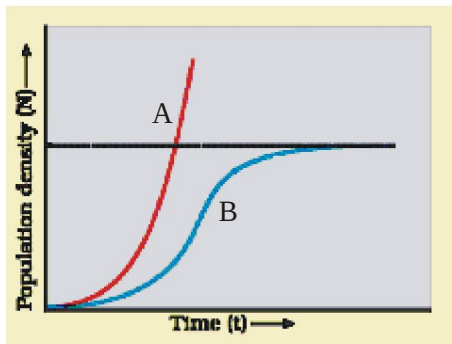
129. Parenchyma and xylem, both are examples of :

- (1) Simple tissue
- (2) Complex tissue
- (3) Permanent tissue
- (4) Secondary meristematic tissue

130. Cell wall of sclerenchymatous fibres and sclereids are respectively thickened by :

- (1) Cellulose, cellulose
- (2) Cellulose, hemicelluloses
- (3) Lignin, cellulose
- (4) Lignin, lignin

131. Identify the A and B in given population growth curve :-



- (1) A growth curve-when resources are limiting
- (2) B growth curve-when resources are not limiting
- (3) A growth curve-when resources are not limiting
- (4) B is exponential growth curve when resource

132. World AIDS day is designated on –

- (1) 1st October
- (2) 1st November
- (3) 1st January
- (4) 1st December

133. The logistic population growth is expressed by the equation :-

$$(1) \frac{dN}{dt} = rN \left[\frac{N - K}{K} \right]$$

$$(2) \frac{dt}{dN} = Nr \left[\frac{K - N}{K} \right]$$

$$(3) \frac{dN}{dt} = rN \left[\frac{K - N}{K} \right]$$

$$(4) \frac{dN}{dt} = rN$$

134. Asymptote in a logistic growth curve is obtained when :-

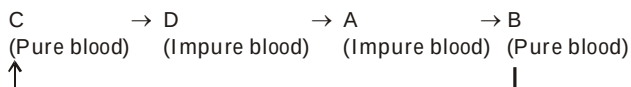
- (1) $K = N$
- (2) $K > N$
- (3) $K < N$
- (4) The value of 'r' approaches zero

135. Which of the following organism is likely to have more concentration of DDT in its body :-

- (1) Primary producers
- (2) Herbivores
- (3) Carnivores
- (4) Top carnivores

136. Which of the following statements is the most correct regarding excitation of heart muscles?
- (1) Only the A.V. nodal musculature has the ability to generate action potential
 - (2) Only the SA node in heart has the ability to generate action potential
 - (3) All the nodal tissue have ability to generate action potential
 - (4) Only the purkinje fibre is auto excitable centre of the heart
137. Which of the following factors is not always reported in RBC?
- (A) Antigen
 - (B) Rh factor
 - (C) Nucleus
 - (D) Haemoglobin
 - (E) Biconcave shape
- (1) (A), (B), (C) and (D)
 - (2) (B), (C), (D) and (E)
 - (3) (C), (D) and (E)
 - (4) (A), (B), (C) and (E)
138. Which of the following is a matching pair?
- (1) Initiation of heart beat – Purkinje fibres
 - (2) Pulsation of radial artery – Valves in blood vessels
 - (3) Dup – opening of semilunar valves at the beginning of ventricular systole
 - (4) Lubb – Closure of AV valves at the beginning of ventricular systole
139. Which of the following statements is not correct about all vertebrate's heart?
- (1) Presence of muscular chambered heart
 - (2) Presence of SA node
 - (3) Presence of myogenic heart
 - (4) It always pumps together oxygenated and deoxygenated blood to the different body parts.
140. Which of the following is correct?
- (1) Nearly 70% of blood is received by ventricles during joint diastole
 - (2) Auricular systole and ventricular systole never overlap
 - (3) The duration of joint diastole is 0.4 seconds.
 - (4) All are correct
141. Total thrombocytic count in human blood is :
- (1) 15,000 - 25,000
 - (2) 1,500,00 - 3,500,00
 - (3) 3,000,00 - 5,000,00
 - (4) 10,000 - 20,000
142. Conversion of fibrinogen in fibrin filaments is very important for blood clotting, which occurs by catalysis of :
- (1) Prothrombinase
 - (2) Thrombokinase
 - (3) Thrombin
 - (4) Prothrombin
143. Which of following is incorrect with respect to human heart ?
- (1) It is tetrachambered and myogenic
 - (2) Surrounded by pericardium
 - (3) Located in mediastinum
 - (4) Endodermal in origin
144. Select a correct statement about working of heart :
- (1) It beats 72 times per minute
 - (2) Pumps 70 ml blood in each aorta by single systole
 - (3) Produces LUBB and DUP sound in each cardiac cycle
 - (4) All of these.
145. Myocardial ischemia is insufficient blood and O_2 supply to heart wall which is initially characterised by:
- (1) Cardiac arrest
 - (2) Angina pectoris
 - (3) Heart failure
 - (4) Hypertension

146. Human heart has four chambers which are labelled as A, B, C & D. The direction of blood flow and its oxygen content is as shows below :-



Here the correct names of A, B, C & D will be :-

- (1) A – Right Atrium; B – Right Ventricle;
C – Left Atrium ; D – Left Ventricle
(2) A – Right Ventricle; B – Left Atrium;
C – Left Ventricle ; D – Right Atrium
(3) A – Left Atrium; B – Left Ventricle;
C – Right Atrium; D – Right Ventricle
(4) A – Left Ventricle; B – Right Atrium;
C – Right Ventricle; D – Left Atrium
147. If six liters of deoxygenated blood enter the lungs via pulmonary artery in one minute time then the amount of oxygenated blood which will exit the lungs via pulmonary veins in a minute time will be :-
- (1) 6 litre (2) < 6 litre
(3) > 6 litre (4) Data insufficient
148. Match the blood vessels of human heart listed under column-I with the functions given under column-II. Choose the answer which gives the correct combination of the alphabets of the two columns.

Column-I (Blood vessel)		Column-II (Function)	
(A)	Superior vena cava	p.	Carries deoxygenated blood to lungs
(B)	Inferior vena cava	q.	Carries oxygenated blood from lungs
(C)	Pulmonary artery	r.	Brings deoxygenated blood from lower parts of the body to the right atrium
(D)	Pulmonary vein	s.	Brings oxygenated blood to the left atrium
		t.	Brings deoxygenated blood from upper parts of the body into the right atrium

- (1) A = t, B = p, C = r, D = q
(2) A = t, B = r, C = p, D = s
(3) A = s, B = t, C = r, D = p
(4) A = t, B = p, C = q, D = r

149. What is the state of heart in the interval after end of second heart sound and before beginning of first heart sound again :-
- (1) Ventricle remain in systole Auricles undergo first diastole and then systole
(2) Ventricle remain in diastole Auricles undergo first diastole and then systole
(3) Ventricles undergo first systole and then diastole Auricles remain in diastole
(4) Ventricles remain in diastole Auricles remain in systole
150. Volume of blood pumped out by each ventricle per minute is called :-
- (1) Stroke volume
(2) Cardiac output
(3) Cardiac index
(4) Tidal volume
151. In ECG, P-wave represents
- (1) Atrial repolarization
(2) Atrial depolarization
(3) Ventricular depolarization
(4) Ventricular repolarization
152. Pulse pressure is
- (1) Systolic pressure
(2) Diastolic pressure
(3) Pressure in vena cava
(4) Difference between systolic and diastolic pressure

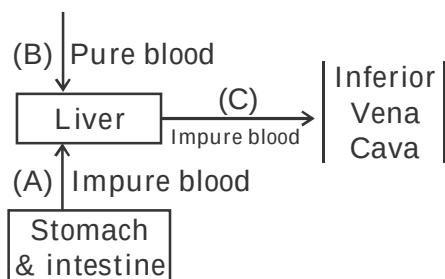
153. The greatest percentage of blood volume is found in

- (1) Veins (2) Arteries
(3) Capillaries (4) Heart

154. In a study of four blood vessels A, B, C & D in man, vessel 'A' carried pure blood away from heart. Vessel 'B' carried impure blood towards heart. Vessel 'C' carried pure blood into the heart musculature while vessel 'D' carried impure blood away from heart. The correct names of the four vessels can be :

- (1) Coronary artery-A, Pulmonary vein-B, Pulmonary Artery-C, SVC-D
(2) Pulmonary vein-A, Pulmonary Artery-B, Coronary vein-C, Coronary Artery-D
(3) Systemic Aorta-A, Vena cava-B, Coronary artery-C, Pulmonary vein-D
(4) Systemic Aorta-A, Vena cava-B, Coronary Artery-C, Pulmonary Artery-D

155. Please observe the diagram below



Here vessel A, B and C are -

- (1) A - Hepatic vein
B - Portal vein
C - Hepatic artery
(2) A - Portal vein
B - Hepatic artery
C - Hepatic vein
(3) A - Hepatic artery
B - Hepatic vein
C - Portal vein
(4) A - Portal vein
B - Hepatic vein
C - Hepatic artery

156. In a human cardiac cycle which of the following statement is false :-

- (1) The time taken for one cardiac cycle decreases in Tachycardia
(2) The time interval for joint Diastole in heart is about 0.4 seconds
(3) The time interval for joint systole in heart is about 0.4 second
(4) Left ventricle takes 0.5 seconds to complete one diastole

157. Consider the following four statements (a-d) and select the option which includes all the correct :-

- (a) Coronary Artery disease. (CAD) often referred to as Atherosclerosis
(b) Heart failure means when the heart muscle is suddenly damaged by an inadequate blood supply
(c) High blood pressure leads to heart diseases and also affects vital organs like brain and kidney
(d) Angina occurs due to conditions that affect the blood flow

Options :-

- (1) Statements (b), (c) and (d)
(2) Statements (a), (b)
(3) Statements (b), (d)
(4) Statement (a), (c) and (d)

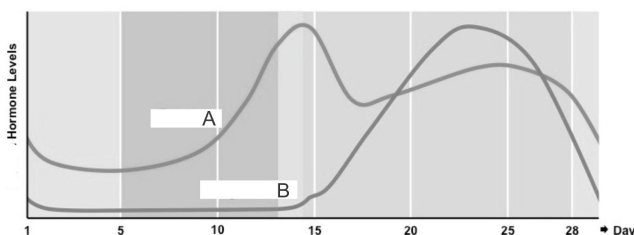
158. Given below is a table showing various blood groups and donor compatibility. Identify the incorrect option :-

	Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
(1)	A	A	anti-B	A, O
(2)	B	B	anti-A	B, O
(3)	AB	A, B	anti-A, B	AB, A, B, O
(4)	O	nil	anti-A, B	O

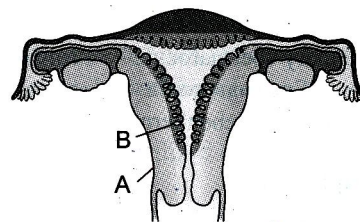
159. Atrioventricular node (AV node) is present in

- (1) Right upper corner of right atrium
(2) Lower left corner of right atrium
(3) Right ventricle
(4) Left atrium

160. Human heart is myogenic because
 (1) Sympathetic nerves can increase heart rate
 (2) Impulse for heart beat is generated in SA node
 (3) Impulse for heart beat is generated outside the heart
 (4) Myocardium is present in heart wall
161. In the foetal heart, ductus arteriosus connects
 (1) Pulmonary artery with pulmonary vein
 (2) Pulmonary artery with systemic aorta
 (3) Right atrium and left atrium
 (4) Pulmonary vein with systemic aorta
162. The exchange of materials between blood and interstitial fluid takes place through
 (1) Veins
 (2) Arteries
 (3) Arterioles
 (4) Capillaries
163. Slowing down of heart rate is called
 (1) Tachycardia
 (2) Bradycardia
 (3) Bradypnoea
 (4) Dextrocardia
164. Heart rate is accelerated by
 (1) Parasympathetic nerves and acetylcholine
 (2) Sympathetic nerves and epinephrine
 (3) Sympathetic nerves and acetylcholine
 (4) Parasympathetic nerves and epinephrine
165. If vagus nerve is stimulates, the heart rate
 (1) Increases
 (2) Decreases
 (3) Remains unchanged
 (4) First increases, then decreases
166. Identify the hormones labelled as A and B in the following graph.



- (1) A - LH, B - Progesterone
 (2) A - Oestrogen, B - FSH
 (3) A - Oestrogen, B - Progesterone
 (4) A - Progesterone, B - Oestrogen
167. In female external genitalia, a pair of large skin folds homologous with scrotum of male is -
 (1) Labia minora
 (2) Labia majora
 (3) Clitoris
 (4) Fourchette
168. Which of the following structure is/are present in vestibule part of female reproductive system?
 (i) Clitoris
 (ii) Perineum
 (iii) Urethral orifice
 (iv) Vaginal orifice
 (1) (i) and (ii) (2) (iii) and (iv)
 (3) (ii) and (iii) (4) (i) and (iv)
169. The cervical canal opens into vagina by an opening called
 (1) Internal Os (2) External Os
 (3) Ampulla (4) Fundus
170. Select the option which gives correct identification of labelled parts A and B along with functional characteristics in the given sectional view of female reproductive system.



- (1) A - perimetrium, composed of ciliated cuboidal epithelium
 (2) B - Myometrium, which undergoes strong uterine contraction at the time of child birth
 (3) B - Endometrium is the inner glandular lining formed of columnar epithelium
 (4) A - Myometrium, made of smooth muscles

171. Difference between oogenesis and spermatogenesis is:

- (1) Spermatogenesis is not completed until after fertilization occurs, but oogenesis is completed at the time of birth
- (2) Oogenesis begins at puberty and spermatogenesis begins during embryonic life
- (3) Primary oocyte forms four haploid ova while primary spermatocyte forms only one haploid sperm
- (4) Cytoplasmic division is unequal in oogenesis and equal in spermatogenesis

172. Menarche is the stage when

- (1) First ejaculation occurs
- (2) First menstrual cycle occurs
- (3) Hymen breaks down
- (4) Parturition takes place

173. Stroma of the mammalian ovary consists of

- (1) Connective tissue
- (2) Nerves
- (3) Vascular tissue
- (4) All of these

174. Corpus luteum is developed from

- (1) Primary follicle
- (2) Ovarian stroma
- (3) Left over follicle after release of ovum
- (4) Ovum

175. Zona pellucida, a glycoprotein covering in mature follicle, is an example of

- (1) Primary egg membrane
- (2) Secondary egg membrane
- (3) Tertiary egg membrane
- (4) None of these

176. Which of the following statement is incorrect about menstruation?

- (1) Around 40 – 80 ml blood loss takes places.
- (2) Menses take place for 3 – 5 days.
- (3) Beginning of menstrual cycle at puberty is called menarche.
- (4) Menstrual fluid can clot easily.

177. The levels of which of the following hormone rises in urine, in females after menopause?

- (1) Oestrogen
- (2) Progesterone
- (3) FSH
- (4) Androgens

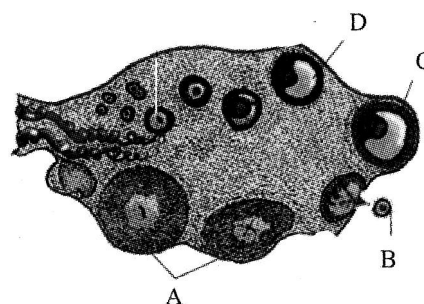
178. Lack of menstruation may be indicative of

- (1) Stress
- (2) Poor health
- (3) Pregnancy
- (4) All of these

179. Antrum, a fluid-filled cavity is typically seen in which follicle?

- (1) Primordial follicle
- (2) Primary follicle
- (3) Secondary follicle
- (4) Graafian follicle

180. The diagram of the sectional view of the ovary is given below. The labelled structure that secretes large amount of progesterone is



- (1) A
- (2) B
- (3) C
- (4) D